Leviathan Prototype Implementation Report

1. Overview

This report documents the capabilities demonstrated in the Leviathan mainnet prototype, evaluates them against the architecture described in the whitepaper, and proposes improvements for further development.

Leviathan is a scaling protocol for Cardano that achieves instant finality by leveraging guaranteed transactions, deterministic transactions with exclusive state access, chained together without waiting for individual L1 confirmations. The prototype validates this architecture on mainnet with a working sequencer, account based intent system, and real time order book.

2. Capabilities Achieved

2.1 Induced Finality via Guaranteed Transaction Chains

The whitepaper's central thesis that guaranteed transactions under assumption of liveness have instant finality is demonstrated in practice. Orders placed through the interface are final immediately upon sequencer inclusion, without waiting for L1 block confirmation. The chain extension mechanism (N+1 transactions consuming outputs of the N-th) operates as theorized, enabling sub-second confirmation of user actions.

2.2 Working Off-Chain Sequencer

A centralized sequencer processes user intents and constructs the Leviathan Chain Extension on mainnet. The sequencer:

- Receives signed intents from users
- Orders them into a deterministic transaction sequence
- Extends the chain in real time
- Has no custody over user funds (at worst, can only halt)

This matches the design where the sequencer holds executive authority to elect blocks in the chain extension while being unable to misappropriate funds.

2.3 Account Abstraction Layer

Users interact through accounts that hold their funds on-chain. The account system:

- Eliminates minimum UTXO requirements for the end user

- Abstracts away gas management
- Enables pre-funded interactions without per transaction UTXO construction
- Decouples the user from the deterministic transaction construction

2.4 Intent-Based Transaction Model

Rather than signing raw L1 transactions, users sign intents; high-level expressions of desired outcomes. The prototype supports:

Intent Type	**Description**
Limit Orders	Place buy/sell orders at a specified price
Order Cancellation	Cancel previously placed orders in real time
Payment	Direct payment to an address
Account Close/Withdraw	Exit the system and retrieve funds to L1

2.5 Real-Time State Updates

The interface reflects state changes immediately upon sequencer confirmation. Order placements, cancellations, and balance updates are visible in real time demonstrating the throughput and latency improvements that Leviathan provides over raw L1 interaction.

2.6 Inverse Rollup Entry/Exit Model

The "time travel" properties described in §3.3 are realized:

- **Entry (L1 > Leviathan):** 12 hours, waiting for L1 finality so the account state becomes guaranteed
- **Exit (Leviathan > L1):** Instant, as funds can be released directly from the on-chain account

This is the inverse of traditional rollups (instant entry, delayed exit) and is a direct consequence of Cardano's finality window combined with the guaranteed transaction model.

2.7 Safety Properties

- The sequencer cannot steal funds, it can only halt
- Self-matching protection is enforced (users cannot match their own orders)
- All intents are cryptographically signed
- Once on the Leviathan ledger, transactions are final and immutable by the sequencer

3. Current Limitations

Limitation	Impact
Centralized sequencer	Single point of liveness failure; no decentralized ordering consensus yet
No wallet integration	Users must interact through a custom signing flow rather than standard Cardano wallets
Prototype fund caps	Limited amounts allowed to constrain risk during testing
No data compression	L1 cost parity. No rollup style savings (acknowledged in whitepaper §3)
No ZK layer	Account state is not compressed; full L1 data footprint per transaction
12-hour entry time	Barrier to onboarding; no fast-bridge mechanism yet

4. Proposed Improvements

4.1 Wallet Integration

Priority: High
The prototype requires a custom signing flow. Integration with wallets (Eternl already in progress) would:

- Enable standard CIP-30 signing of intents
- Present decoded intent data to users for safe verification
- Lower friction for onboarding existing Cardano users

Approach: Define a CIP standard message format for Leviathan intents that wallets can parse and display meaningfully.

4.2 Fast Entry Bridge

Priority: High
The 12-hour entry time is a significant UX barrier. A liquidity backed bridge could allow:

- Instant entry by fronting funds from a liquidity pool
- The bridge operator assumes finality risk in exchange for a fee
- Similar to how L2 bridges on Ethereum provide instant exits from rollups

4.3 Sequencer Decentralization

Priority: Medium
The whitepaper acknowledges this as the hardest open problem (§3.2). A phased approach:

- **Phase 1 (Current):** Single trusted sequencer
- **Phase 2:** Multi-sequencer with rotating leadership (round robin or stake weighted)
- **Phase 3:** BFT consensus among sequencer set with cryptographic proof of correct ordering passed to the L1 chain extension

Each phase increases liveness guarantees without changing the fundamental safety property (sequencer cannot steal, only halt).

4.4 Merkle Tree Airdrops

Priority: Medium
Mentioned in the demo as a near-term addition. Implementation would:

- Commit a Merkle root to the Leviathan ledger
- Allow instant claims via Merkle proof verification within the account framework
- Demonstrate extensibility of the intent system beyond trading

4.5 Zero-Knowledge State Compression

Priority: Medium
The whitepaper identifies ZK techniques as the path to solving Leviathan's cost problem:

- Compress account state transitions into ZK proofs
- Reduce L1 data footprint per chain extension
- Move toward a ZK rollup architecture while retaining the guaranteed transaction safety model

This would address the economic limitation of L1 parity costs.

4.6 Expanded Application Framework

Priority: Medium
The account based execution layer can support applications beyond order books:

- **Lending/Liquidations:** Instant liquidation execution without L1 latency
- **Oracle Integration:** Real time price feeds delivered within the Leviathan context
- **Programmable Accounts:** Autonomous transaction logic (limit orders are the first example; generalizable to stop losses, DCA, conditional transfers)
- **Token Standards:** Full ERC-20 equivalent fungible token support within the framework

5. Architectural Alignment Summary

Whitepaper Concept	**Prototype Status**	**Next Step**
Guaranteed Transactions	Implemented	Formal verification
Induced Finality	Demonstrated on mainnet	Performance benchmarking
Account Abstraction	Working (accounts, gas, pre-funding)	Wallet integration
Sequencer Network	Single centralized sequencer	Multi-party consensus
Intent System	Limit orders, cancellations, payments	Generalized intent grammar
ZK Compression	Not implemented	Research phase
Application Framework	Order book	Lending, oracles, programmable accounts

6. Conclusion

The Leviathan mainnet prototype validates the core theoretical contributions of the whitepaper: guaranteed transactions enable instant finality on Cardano without sacrificing the security properties of the UTXO model. The implementation demonstrates that the account based intent architecture successfully decouples users from L1 transaction mechanics while maintaining a non-custodial safety guarantee.

The primary vectors for advancement are (1) reducing entry friction through bridging, (2) decentralizing the sequencer for liveness guarantees, and (3) introducing ZK compression for economic viability at scale. Each of these extends the existing architecture without requiring fundamental redesign, evidence that the underlying theoretical framework is sound and production viable.